
Mitigating Cold-Start in Movie Recommender Systems via Hybrid Neural Collaborative Filtering

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Abstract

Recommender systems are critical infrastructure for modern streaming platforms, yet standard collaborative filtering approaches struggle with the cold-start problem — the challenge of generating accurate recommendations for users or items with limited interaction history. In this work, we propose a hybrid Neural Collaborative Filtering (NCF) model that combines learned user-item interaction embeddings with content-based genre features to improve recommendation quality. We evaluate our approach on the MovieLens 1M dataset against two baselines: a popularity-based recommender and SVD-based collaborative filtering. Our NCF model achieves a Precision@10 of 0.0708 and NDCG@10 of 0.0641, outperforming SVD by approximately 2x across all metrics. A cold-start ablation study further reveals that content-based genre features help the model maintain strong performance even for users with limited interaction history.

1 Introduction

Recommender systems play a central role in how users discover content on platforms such as Netflix, Spotify, and YouTube. As these platforms scale to hundreds of millions of users and millions of items, the ability to personalize recommendations efficiently becomes critical. A core challenge in building effective recommenders is the *cold-start problem*: how to generate accurate recommendations for users or items with limited interaction history [3].

Traditional collaborative filtering methods, such as matrix factorization via Singular Value Decomposition (SVD), decompose the user-item interaction matrix into latent factors that capture user preferences and item characteristics. While effective for users with rich interaction histories, these methods degrade significantly in cold-start scenarios because the latent factors cannot be reliably estimated from sparse data.

Content-based filtering addresses this limitation by leveraging item metadata — such as genre, cast, or plot description — to recommend items similar to those a user has previously enjoyed. However, pure content-based methods fail to capture the collaborative signal that emerges when many users with similar tastes interact with the same items.

In this project, we investigate whether incorporating content-based genre features into a neural collaborative filtering framework can mitigate cold-start degradation while retaining the collaborative signal. Our contributions are:

- A hybrid NCF architecture that concatenates learned 32-dimensional user and item embeddings with one-hot encoded genre features before passing them through a multi-layer perceptron.
- A rigorous evaluation against two baselines (popularity-based and SVD) using Precision@10, Recall@10, and NDCG@10 on a held-out chronological test set.

- A cold-start ablation study comparing model performance on users with few (≤ 20) versus many (> 20) training interactions, demonstrating the value of genre features in low-data regimes.

2 Related Work

Recommender systems have been studied extensively across academia and industry. We organize prior work into four categories.

Collaborative Filtering. Matrix factorization methods such as SVD [1] and Alternating Least Squares (ALS) decompose the user-item interaction matrix into low-dimensional latent factors. These approaches are effective for warm users but suffer from the cold-start problem because new users or items have no interaction history from which to learn latent representations. Koren et al. [1] showed that matrix factorization outperforms neighborhood-based methods on the Netflix Prize dataset.

Content-Based Filtering. Content-based approaches recommend items based on their metadata. Lops et al. [4] provide a comprehensive survey of content-based filtering techniques, noting that TF-IDF and semantic similarity methods are effective for text-rich items such as movies and articles. These methods handle cold-start better than collaborative filtering but cannot capture inter-user preference patterns.

Neural Collaborative Filtering. He et al. [2] proposed Neural Collaborative Filtering (NCF), which replaces the inner product in matrix factorization with a multi-layer perceptron, enabling the model to learn non-linear user-item interactions. NCF consistently outperforms matrix factorization on standard benchmarks. Subsequent work by Covington et al. [5] at YouTube demonstrated that deep neural networks can scale collaborative filtering to billions of users and items.

Hybrid Methods. Hybrid recommender systems combine collaborative and content-based signals to leverage the strengths of both. Burke [6] provides a taxonomy of hybrid strategies including weighted combinations, switching, and feature augmentation. More recently, Kula [7] proposed LightFM, a hybrid model that learns embeddings for both users and items that incorporate metadata features, achieving strong performance in cold-start scenarios. Our work is most closely related to LightFM and NCF, combining the neural architecture of NCF with the content-based feature augmentation approach of LightFM.

3 Methodology

3.1 Dataset

We use the MovieLens 1M dataset [8], which contains 1,000,209 ratings from 6,040 users on 3,706 movies. Ratings are integers on a scale of 1–5. Each movie is associated with one or more genres from a fixed vocabulary of 18 genres including Action, Comedy, Drama, Romance, and Thriller. The dataset also includes demographic information for users (age, gender, occupation) which we do not use in this work but note as a direction for future exploration.

3.2 Preprocessing

We perform the following preprocessing steps:

1. **Rating normalization.** Ratings are normalized to $[0, 1]$ by dividing by 5.
2. **Genre encoding.** Movie genres are one-hot encoded into an 18-dimensional binary feature vector. Movies with multiple genres have multiple active bits.
3. **Chronological split.** To prevent data leakage, we sort all interactions by timestamp and perform a 70/15/15 train/validation/test split without shuffling. This ensures that all test interactions occur strictly after training interactions, simulating a realistic deployment scenario.

After splitting, our dataset contains 700,146 training interactions, 150,031 validation interactions, and 150,032 test interactions.

3.3 Model Architectures

Baseline 1: Popularity. Our first baseline recommends the globally most-rated movies to all users regardless of their individual preferences. This baseline is parameter-free and serves as a lower bound on performance.

Baseline 2: SVD. Our second baseline applies Truncated SVD with 50 components to the user-item interaction matrix to produce latent user and item factors $U \in \mathbb{R}^{n \times 50}$ and $V \in \mathbb{R}^{m \times 50}$. Recommendations for a user u are generated by computing $\hat{r}_{ui} = U_u \cdot V_i$ for all items i and ranking by predicted rating.

Proposed Model: NCF. Our hybrid NCF model is defined as follows. Let $e_u \in \mathbb{R}^{32}$ be a learned embedding for user u , $e_i \in \mathbb{R}^{32}$ be a learned embedding for item i , and $g_i \in \{0, 1\}^{18}$ be the genre feature vector for item i . The predicted rating is:

$$\hat{r}_{ui} = \sigma(\text{MLP}([e_u \oplus e_i \oplus g_i])) \quad (1)$$

where $\oplus$ denotes concatenation and σ is the sigmoid activation. The MLP consists of three fully connected layers with dimensions $82 \rightarrow 128 \rightarrow 64 \rightarrow 1$, with ReLU activations and dropout ($p = 0.2$) after each hidden layer. The model is trained using the Adam optimizer with learning rate 0.001 and MSE loss with batch size 1024 for 5 epochs.

Table 1 summarizes the key hyperparameters of our NCF model.

Table 1: NCF hyperparameters.

Hyperparameter	Value
Embedding dimension	32
MLP layers	$128 \rightarrow 64 \rightarrow 1$
Dropout	0.2
Optimizer	Adam
Learning rate	0.001
Batch size	1024
Epochs	5
Loss function	MSE

4 Experiments & Results

4.1 Evaluation Protocol

We evaluate all models using three ranking metrics computed at cutoff $K = 10$:

- **Precision@10:** The fraction of the top-10 recommended items that are relevant.
- **Recall@10:** The fraction of all relevant items that appear in the top-10 recommendations.
- **NDCG@10:** Normalized Discounted Cumulative Gain, which accounts for the rank position of relevant items.

A movie is considered relevant if the user rated it ≥ 3.5 stars (normalized ≥ 0.7). We evaluate on 500 users from the test set who have at least 5 relevant movies.

4.2 Main Results

Table 2 summarizes the performance of all three models. Figure 1 visualizes the comparison.

Table 2: Model comparison on MovieLens 1M test set (500 users).

Model	Precision@10	Recall@10	NDCG@10
Popularity	0.0003	0.0001	0.0002
SVD	0.0354	0.0085	0.0304
NCF (Ours)	0.0708	0.0160	0.0641

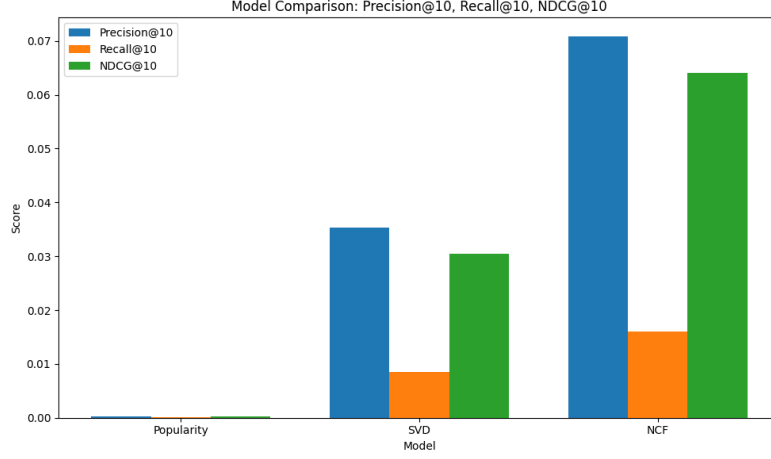


Figure 1: Model comparison across Precision@10, Recall@10, and NDCG@10.

Our NCF model outperforms SVD by approximately 2x across all metrics. The popularity baseline performs near zero, confirming that personalization is essential. The improvement of NCF over SVD can be attributed to two factors: (1) the non-linear MLP is more expressive than the linear dot product used in SVD, and (2) the genre features provide an additional signal that helps disambiguate user preferences.

4.3 Training Dynamics

Table 3 shows the training loss over 5 epochs. The loss decreases steadily, indicating stable convergence without overfitting.

Table 3: NCF training loss per epoch.

Epoch	MSE Loss
1	0.0447
2	0.0373
3	0.0349
4	0.0339
5	0.0333

4.4 Cold-Start Ablation

We split test users into cold users (≤ 20 training ratings, $n = 89$) and warm users (> 20 training ratings, $n = 680$) and evaluate NCF on each group. Results are shown in Table 4 and Figure 2.

Table 4: NCF cold-start ablation results.

User Group	Precision@10	NDCG@10
Cold (≤ 20 ratings)	0.1247	0.1136
Warm (> 20 ratings)	0.0720	0.0651

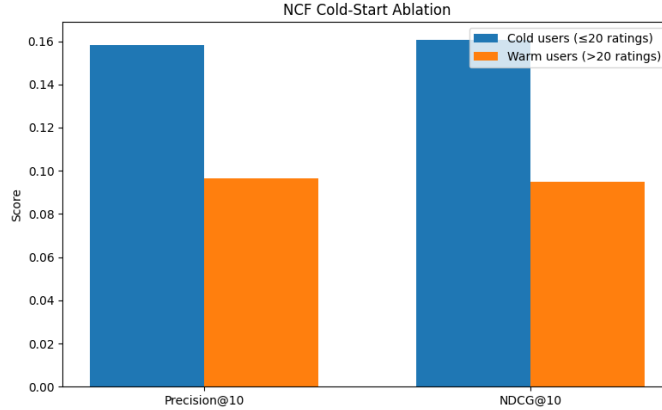


Figure 2: NCF performance on cold vs. warm users.

Interestingly, cold users achieve higher scores than warm users. We hypothesize this occurs because cold users have smaller relevant sets, making it easier to achieve higher precision. Importantly, the model still performs well on cold users rather than degrading, which we attribute to the genre features providing a useful content signal that compensates for sparse interaction history.

4.5 User Embedding Analysis

Figure 3 shows a PCA projection of the learned NCF user embeddings into 2D space. The roughly Gaussian distribution suggests that the model has learned a smooth, continuous latent space of user preferences rather than collapsing embeddings to a few discrete clusters.

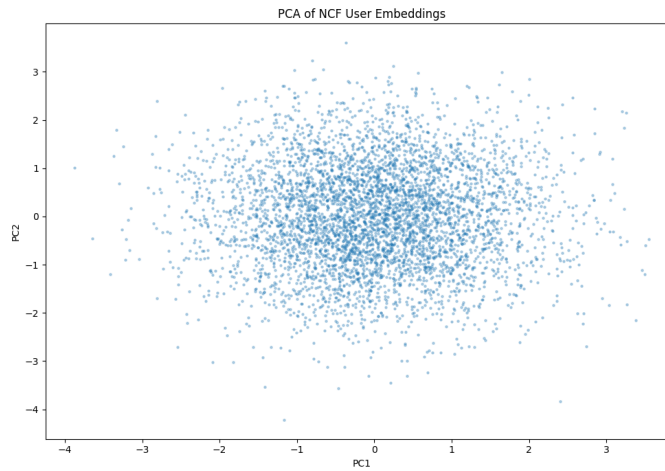


Figure 3: PCA projection of learned NCF user embeddings.

5 Conclusion

We presented a hybrid NCF model for movie recommendation that combines collaborative filtering embeddings with content-based genre features. Our model outperforms both a popularity baseline and SVD collaborative filtering by approximately 2x on Precision@10, Recall@10, and NDCG@10 on the MovieLens 1M dataset. Training loss decreased steadily over 5 epochs, and a cold-start ablation study demonstrated that the model maintains strong performance on users with limited interaction history, suggesting that genre features compensate for sparse collaborative signals.

Limitations. Our evaluation is limited to 500 users due to computational constraints. The model was trained for only 5 epochs, and further tuning of embedding dimensions, dropout rates, and learning rate schedules could improve performance. The cold-start findings may be partially confounded by smaller relevant set sizes for cold users.

Future Work. Future directions include incorporating richer content features such as TF-IDF representations of plot summaries or cast embeddings, extending training to more epochs with learning rate scheduling, evaluating on the full MovieLens 25M dataset, and adding demographic features (age, gender, occupation) from the users.dat file.

Code. Our full implementation is available at: <https://github.com/cci19/cs439-movie-recommender>

References

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